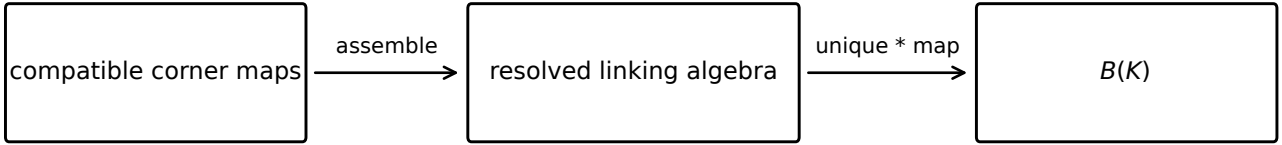
$$p_y A p_x \longmapsto Q_y B(K) Q_x$$



composition and adjoint are checked corner by corner